

Homework 2

Security Governance a.y. 24-25

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January, 2025



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1 Exporting Security Governance to a 3rd World Country

Security governance is a fundamental component of an organization's ability to protect its information technology infrastructure and data assets.

Developing countries face significant challenges in establishing robust IT security frameworks due to limited financial resources, a lack of skilled personnel, and insufficient infrastructure. As these nations undergo rapid digital transformation, they become increasingly vulnerable to cyber threats, ranging from data breaches to ransomware attacks. In the absence of structured security governance, the risks to critical systems, personal data, and economic stability escalate. For these countries, establishing security governance frameworks can safeguard national security, support economic growth, and build trust in digital systems.

1.1 Security Governance

Security governance in IT refers to the frameworks through which organizations control and direct security policies, processes, and technologies. It involves defining roles, responsibilities, and accountability mechanisms to ensure information security aligns with organizational goals. Some of the key concepts to pay attention to include:

- **Accountability:** Clear assignment of responsibilities.
- **Risk Management:** Identifying, assessing, and mitigating risks.
- **Compliance:** Adhering to regulations and standards.
- **Continuous Improvement:** Regular updates to address evolving threats.

Security policies and standards, such as ISO 27001 and the NIST Cybersecurity Framework, act as benchmarks for effective governance. These standards define best practices, ensuring consistent implementation and evaluation of security measures.

1.2 Challenges in Developing Countries

Implementing security governance frameworks in developing nations poses several challenges, for example: limited infrastructure and resources, with outdated hardware, lack of funding, and unreliable internet connectivity that interfere implementation; low awareness of cybersecurity risks and the underestimation of the severity of cyber threats; the lack of skilled and trained personnel and IT security professionals, that delays effective governance.

1.3 Building Awareness and Organization

To address these challenges, initial efforts must focus on specific aspects. For example, community training programs and awareness campaigns are fundamental to educate users about threats and security practices, while establishing basic governance structures are needed to create foundational policies for access control, data protection, and incident response. In some specific contexts, even International Organizations and NGOs are involved to help with the needed expertise and funding to build capacity.

1.4 Proposed Framework for Security Governance

A practical framework for developing countries should prioritize:

- **Core Policies:** Acceptable use policies, access controls, and incident response plans.
- **Compliance Standards:** Adoption of ISO 27001, NIST Framework, or regional equivalents. Given the assumed lack of pre-existing structures, it is beneficial to directly engage with global frameworks in order to be immediately competitive in the global market, avoiding potential future update costs.
- **Tools and Technologies:** Deploying firewalls, antivirus programs, and data encryption solutions as immediate first line of defense.



1.5 Limitations and Realistic Expectations

Especially due to resource constraints, implementation must be phased, starting with a gradual implementation of policies in the most critical security areas and initially addressing only the most critical threats (so prioritizing high-impact vulnerabilities while planning for long-term improvements). This phased approach tries to balance immediate needs with sustainable growth, laying the foundation for more advanced governance models in the future.

2 Prioritizing Security Measures (Case Study)

When it comes to security governance in developing countries, one of the most complex challenges is determining where to begin. The constraints (financial, infrastructural, or human resources) make it impossible to address every security concern simultaneously. This is where prioritization becomes essential, allowing decision-makers to focus on the most pressing vulnerabilities first while planning for gradual enhancements over time.

To better understand this process, let us consider a hypothetical case study of a small government IT network within a developing nation. The network in question supports administrative functions, such as public records management and communication systems, and has recently faced multiple instances of malware infections and unauthorized access attempts. In this scenario, the absence of formalized policies has left the network highly vulnerable to disruptions and breaches.

Analyzing the current state (AS-IS state) of this network reveals a lack of basic controls, such as password policies, firewalls, and incident response plans. Most employees are unaware of cybersecurity best practices, and there are no established protocols for handling security incidents. The assessment highlights several risks, including data loss, reputational damage, and potential legal liabilities.

The vision for the desired state (TO-BE state) involves establishing foundational policies and implementing essential security controls. The first step in achieving this transformation is to prioritize actions based on urgency and impact. For instance, enforcing access controls and setting up network monitoring systems should be addressed immediately, as these measures can significantly reduce vulnerabilities with minimal investment.

Short-term priorities may include developing an incident response plan and implementing regular backup procedures. These steps ensure that even in the event of a breach, data can be recovered, and operations can resume quickly. Longer-term goals, such as deploying advanced threat detection tools and formalizing cloud security policies, provide scalability and resilience as the organization matures.

The success of this transition depends on continuous improvement. Regular assessments, feedback from stakeholders, and training programs help reinforce security practices and adapt to emerging threats. Metrics like reduced incident frequency, improved compliance rates, and staff participation in awareness programs can track progress and demonstrate the effectiveness of the governance framework.

2.1 Key Performance Indicators (KPIs) and Feedback Loops

Measuring the effectiveness of security governance requires well-defined Key Performance Indicators (KPIs). KPIs are a collection of quantifiable measurements, used to assess the overall long-term performance of a business. These may include the number of incidents detected and mitigated, compliance with established policies, response times to incidents, and improvements in employee awareness. Regular performance evaluations enable organizations to track progress and make data-driven adjustments.

Feedback loops further support continuous improvement. Routine audits, penetration testing, and stakeholder feedback help refine security measures. Establishing mechanisms for reporting issues and reviewing incidents ensures that policies remain relevant and effective as new threats emerge.

This case study underscores that prioritization is not just about addressing immediate risks but also about laying the groundwork for sustainable growth. By balancing short-term actions with long-term planning, developing nations can build security governance models that are both resilient and adaptable.

3 Conclusions

Implementing security governance frameworks in developing countries offers both challenges and opportunities. While limited resources and infrastructure require prioritization, starting with foundational policies and gradually expanding capabilities can



deliver meaningful progress. The key to success lies in balancing short-term actions with long-term strategies, supported by clear KPIs and feedback mechanisms.

The adoption of international standards and frameworks provides a roadmap for consistency and reliability, while localized adaptations ensure the approach remains contextually relevant. Emphasizing continuous improvement through KPIs and feedback loops creates a culture of accountability and adaptability, enabling developing nations to scale their security capabilities as resources and knowledge expand.

In conclusion, security governance must be treated as a dynamic process rather than a one-time implementation. With ongoing evaluations, partnerships, and capacity-building efforts, developing countries can create resilient, scalable frameworks capable of addressing present challenges and future uncertainties in cybersecurity.